

Capture The Flag

Mission: Learn how Angular applications are built using reusable components.

Objective: Transform your first Angular application into a modular application using reusable components while following a basic Git workflow.

Flag 0 – Return to Base

Mission

Open the Angular project you created during the previous activity.

Tasks

- Open **Visual Studio Code**.
- Open your Angular project folder.
- Open the integrated terminal.
- Verify that the project still runs.
 - ng serve

Open:

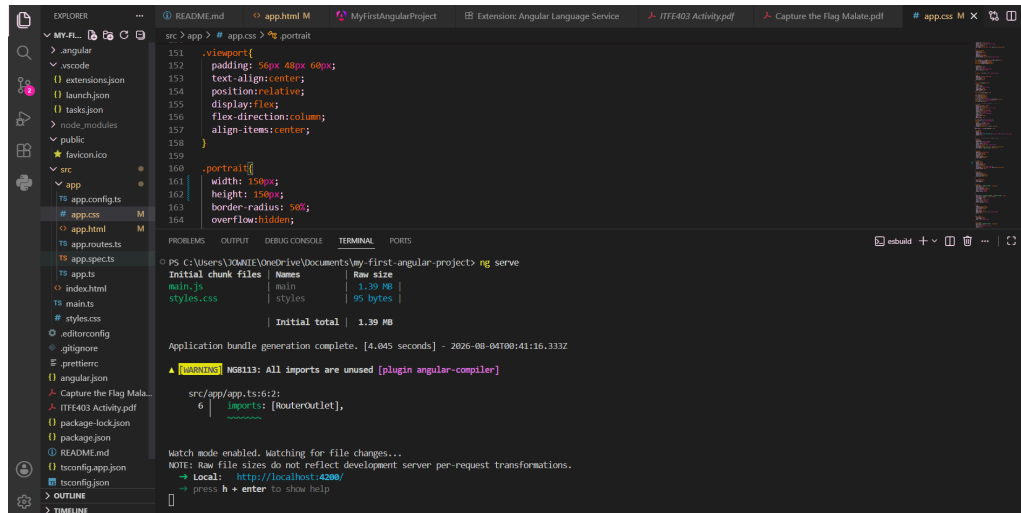
`http://localhost:4200`

If your Angular welcome page appears, proceed.



Evidence

- VS Code with project opened
- Terminal showing `ng serve`
- Browser displaying the Angular application



Flag 1 – Mission Branch

Mission

Every new feature must have its own Git branch.

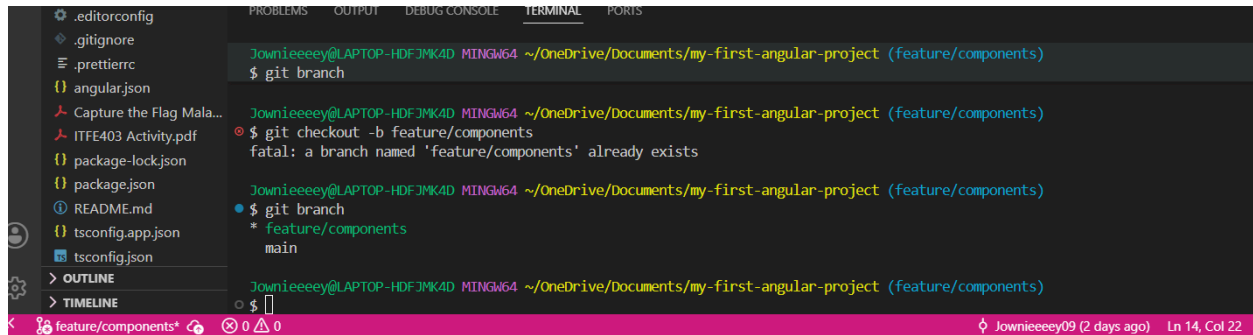
Create a new branch named:

feature/components

Commands:

Evidence

- Terminal showing the successful branch creation
- Output of `git branch`



The screenshot shows a VS Code interface with a terminal window open. The terminal is running the following commands and output:

```
Jownieeey@LAPTOP-HDFJMK4D MINGW64 ~/OneDrive/Documents/my-first-angular-project (feature/components)
$ git branch

Jownieeey@LAPTOP-HDFJMK4D MINGW64 ~/OneDrive/Documents/my-first-angular-project (feature/components)
$ git checkout -b feature/components
fatal: a branch named 'feature/components' already exists

Jownieeey@LAPTOP-HDFJMK4D MINGW64 ~/OneDrive/Documents/my-first-angular-project (feature/components)
$ git branch
* feature/components
main
```

The status bar at the bottom indicates the current branch is 'feature/components' and the file being edited is 'app.html' at line 14, column 22.

Flag 2 – Explore the Base

Mission

Investigate your Angular project before making changes.

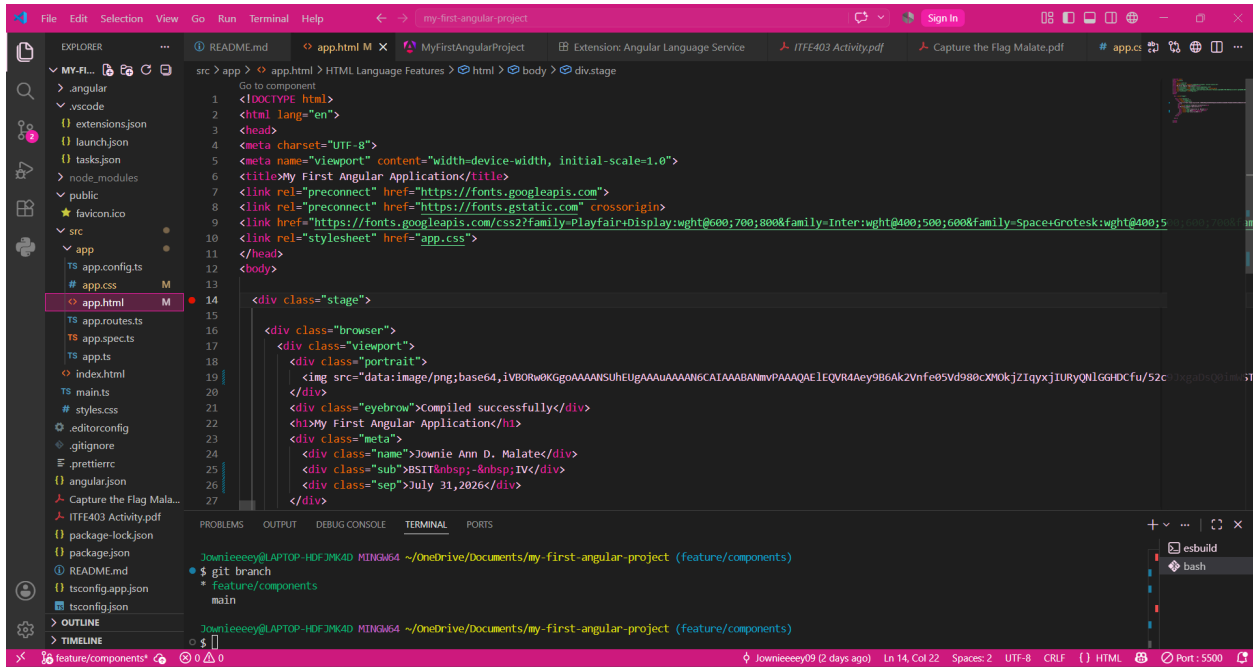
Locate the following:

- `app.html`
- `app.ts`
- `app.css`
- `app.spec.ts`

Understand that these files belong to the **App Component**, the root component of the application.

Evidence

- Explorer showing the files
- `app.html` opened



Flag 3 – Clear the Battlefield

Mission

Remove the default Angular welcome page.

Delete everything inside:

app.component.html

Replace it with:

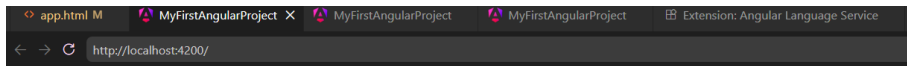
```
<h1>My Component Factory</h1>
```

Verify the browser updates.



Evidence

- Browser showing only the heading
- HTML file



My Component Factory

Flag 4 – Build the Headquarters

Mission

Create a reusable Header component.

Generate:

```
ng generate component components/header
```

Angular should automatically generate:

header.ts

header.html

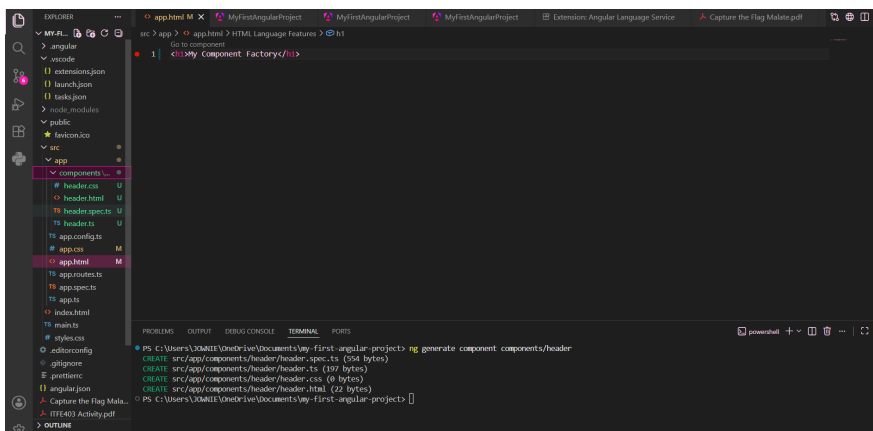
header.css

header.spec.ts



Evidence

- Terminal
- Explorer showing generated files



Flag 5 – Raise the Banner

Mission

Inside the Header component, create a simple header.

Example:

```
<header>
  <h2>Don Bosco Student Portal</h2>
</header>
```

Use the component inside:

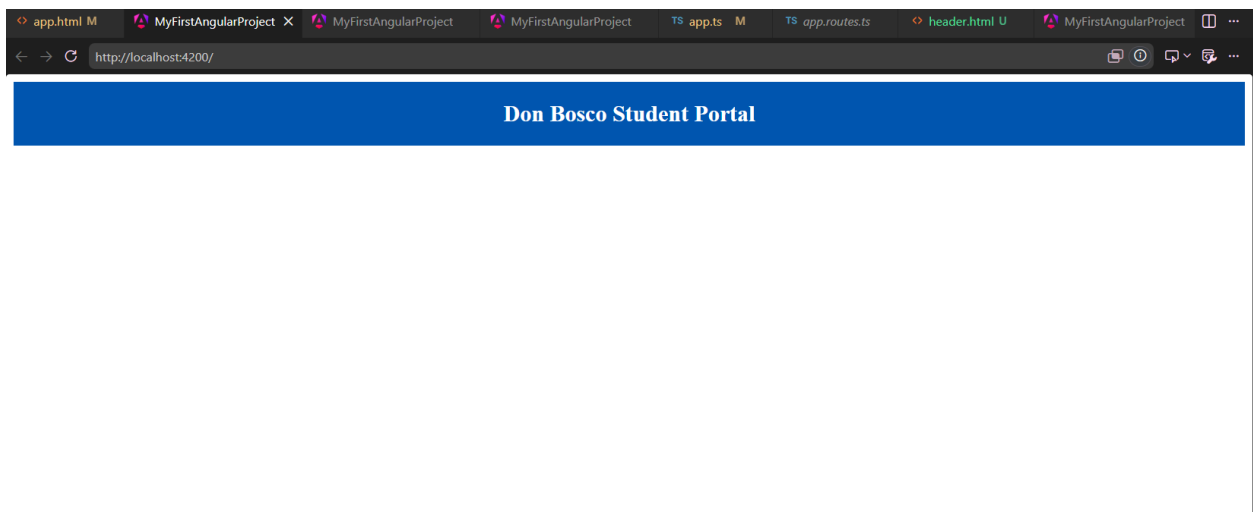
```
app.html
<app-header></app-header>
```

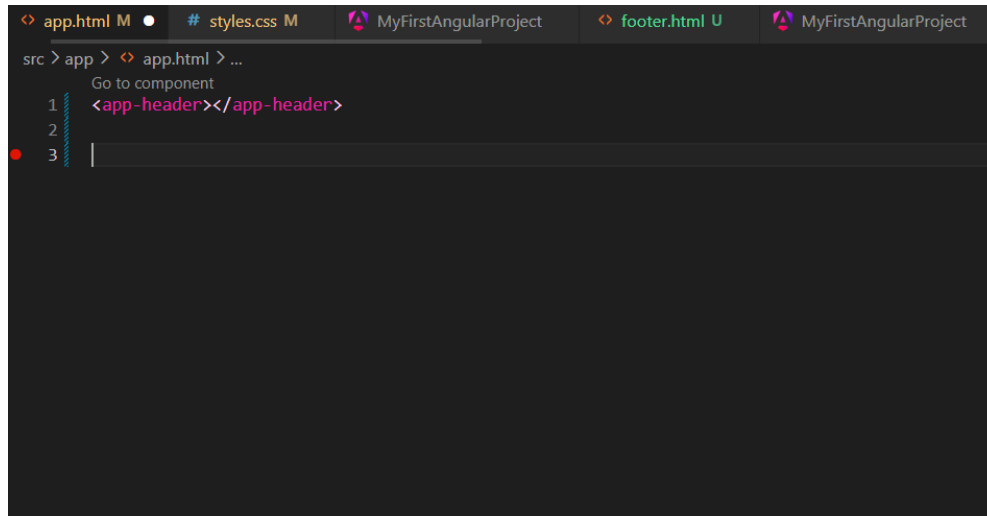
Verify it appears in the browser.



Evidence

- Browser
- HTML code





```
src > app > < app.html > ...
Go to component
1 <app-header></app-header>
2
3 |
```

Flag 6 – Secure the Exit

Mission

Create a Footer component.

Use the same command previously, but now for the footer.

Display:

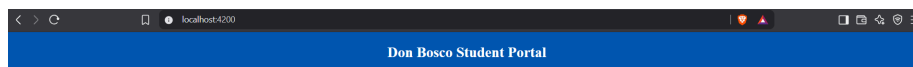
```
<footer>
  © 2026 Don Bosco Technical College
</footer>
```

Insert it into the App Component.



Evidence

- Browser
- Generated files



Flag 7 – Create the Profile Card

Mission

Generate a reusable Profile Card.

Use the command to create a new component called profile-card.

Create a card containing:

- Profile Picture
- Name
- Course
- Year Level

(Design is optional)

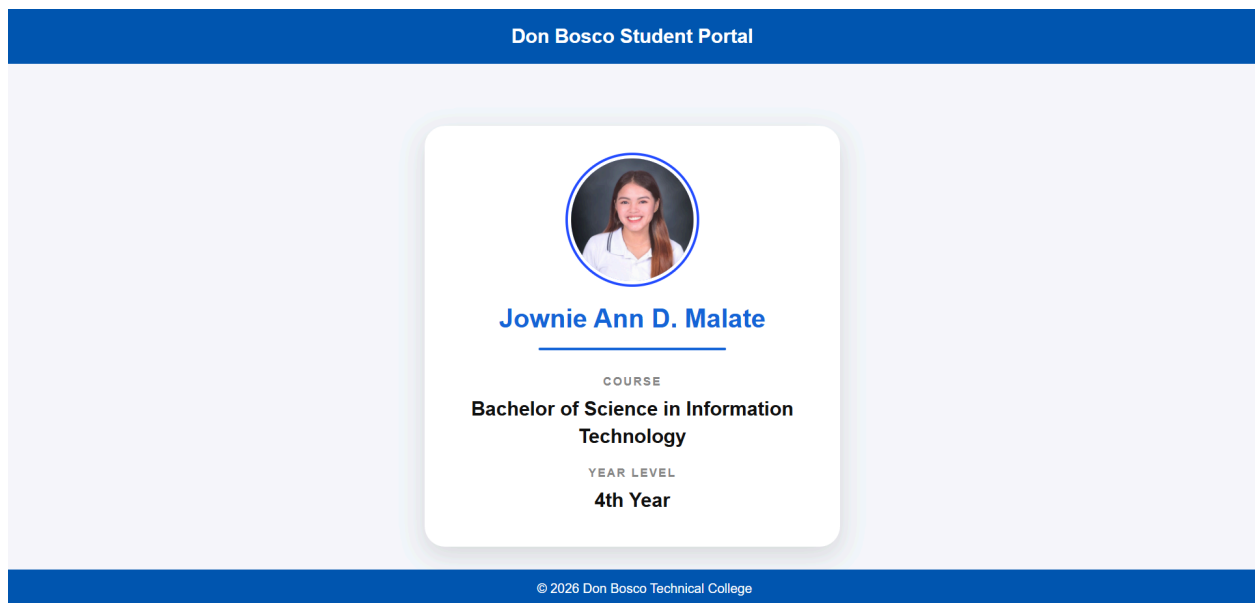
Insert it into:

app.html



Evidence

- Browser
- Profile Card component



Flag 8 – Produce Announcement Cards

Mission

Generate one reusable Announcement Card.

Use the same command, creating the component announcement-card.

Design a card containing:

- Announcement Title
- Description
- Date

Do **NOT** create three separate components.

Instead, reuse the same component.

Example:

```
<app-announcement-card></app-announcement-card>
```

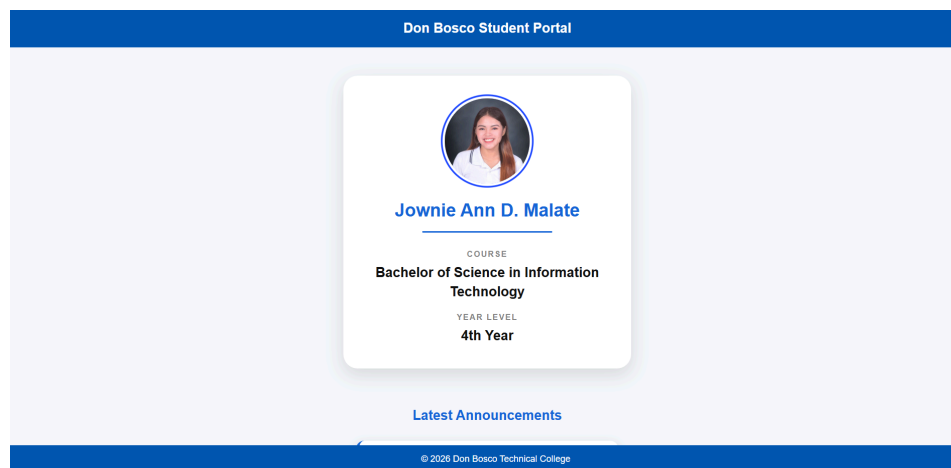
```
<app-announcement-card></app-announcement-card>
```

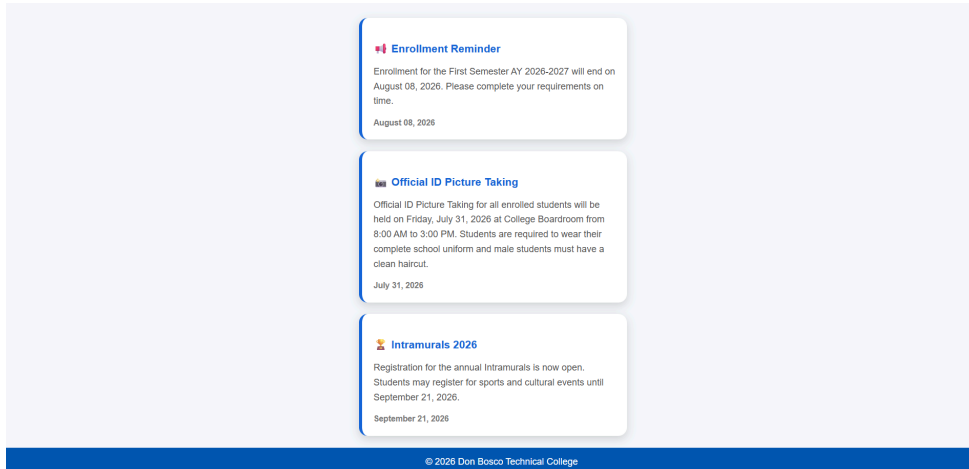
```
<app-announcement-card></app-announcement-card>
```



Evidence

- Browser
- App component showing three announcement cards





Flag 9 – Assemble the Dashboard

Your `app.html` should now act only as the page assembler.

Assemble your page to have this structure:

header

profile-card

announcement-card

announcement-card

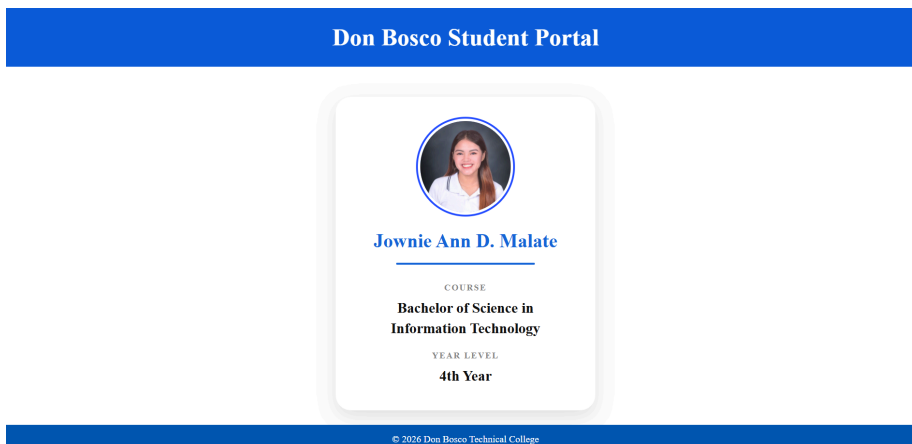
announcement-card

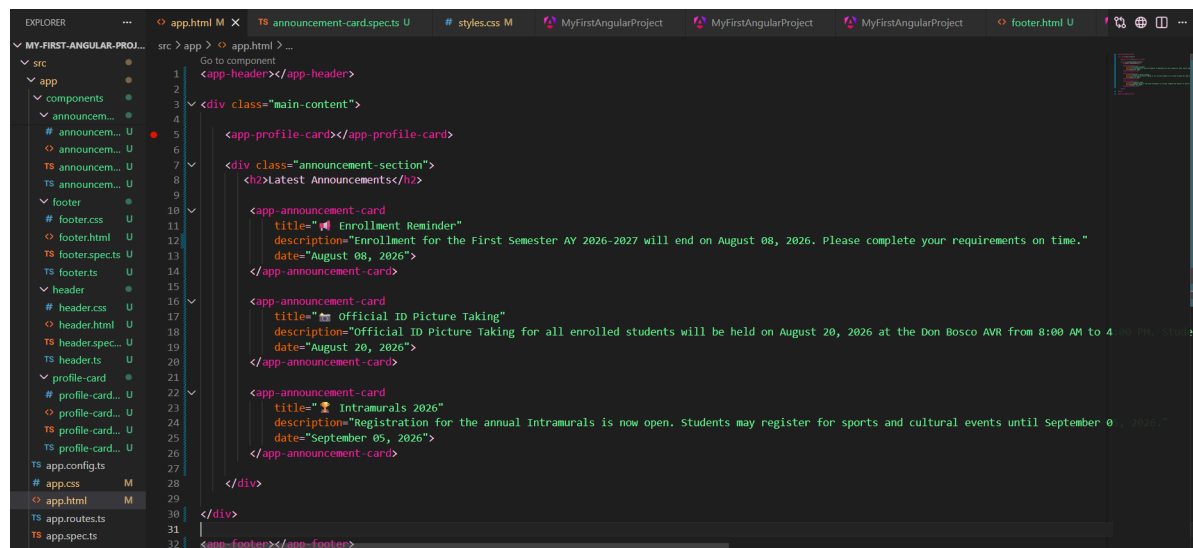
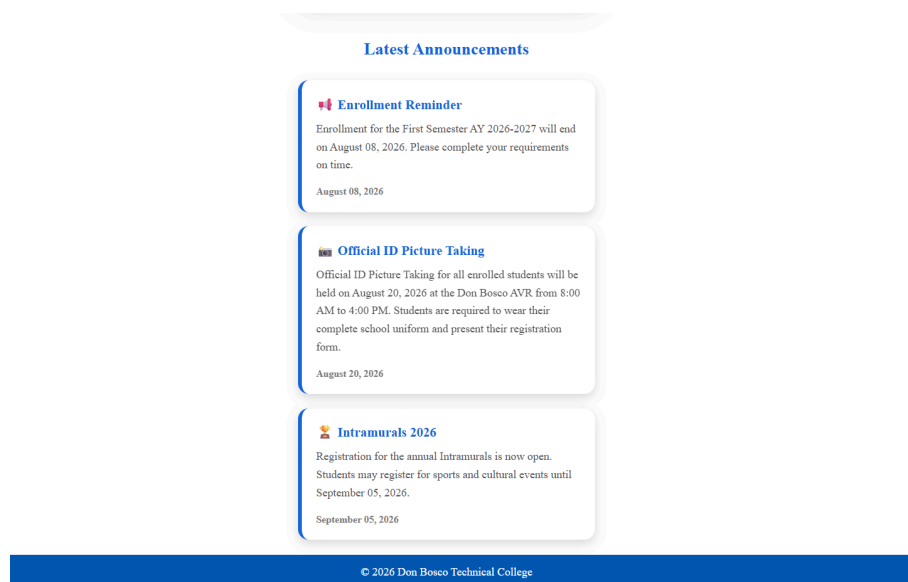
footer

Notice that almost no actual page content exists here. Only reusable components.

Evidence

- Browser
- `app.html`





Flag 10 – Client Changed Their Mind

Mission

The client sends an urgent request:

"Every announcement card should now have:

- Rounded corners
- A blue border
- Padding

- A subtle shadow"

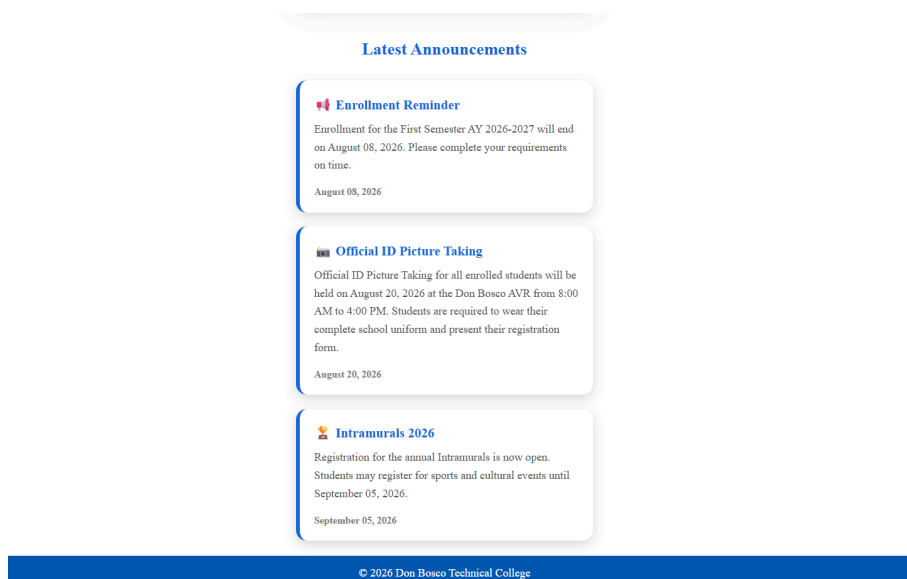
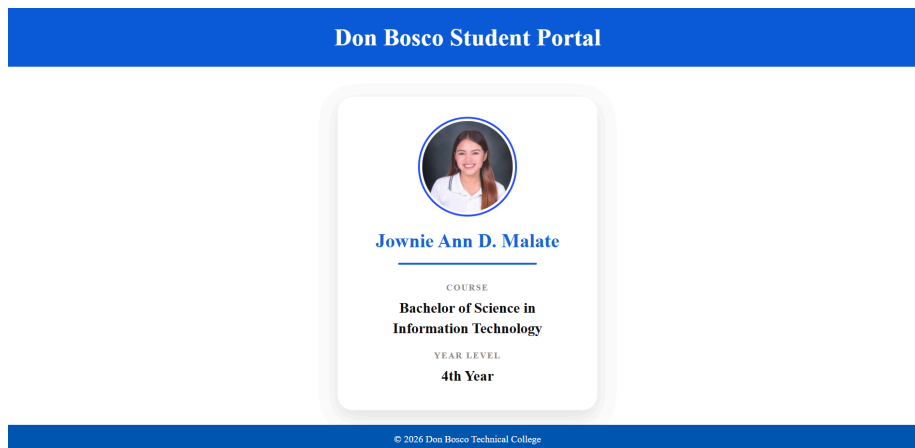
You must update **only one file**.

Apply your styles there and verify that **all announcement cards** update automatically.

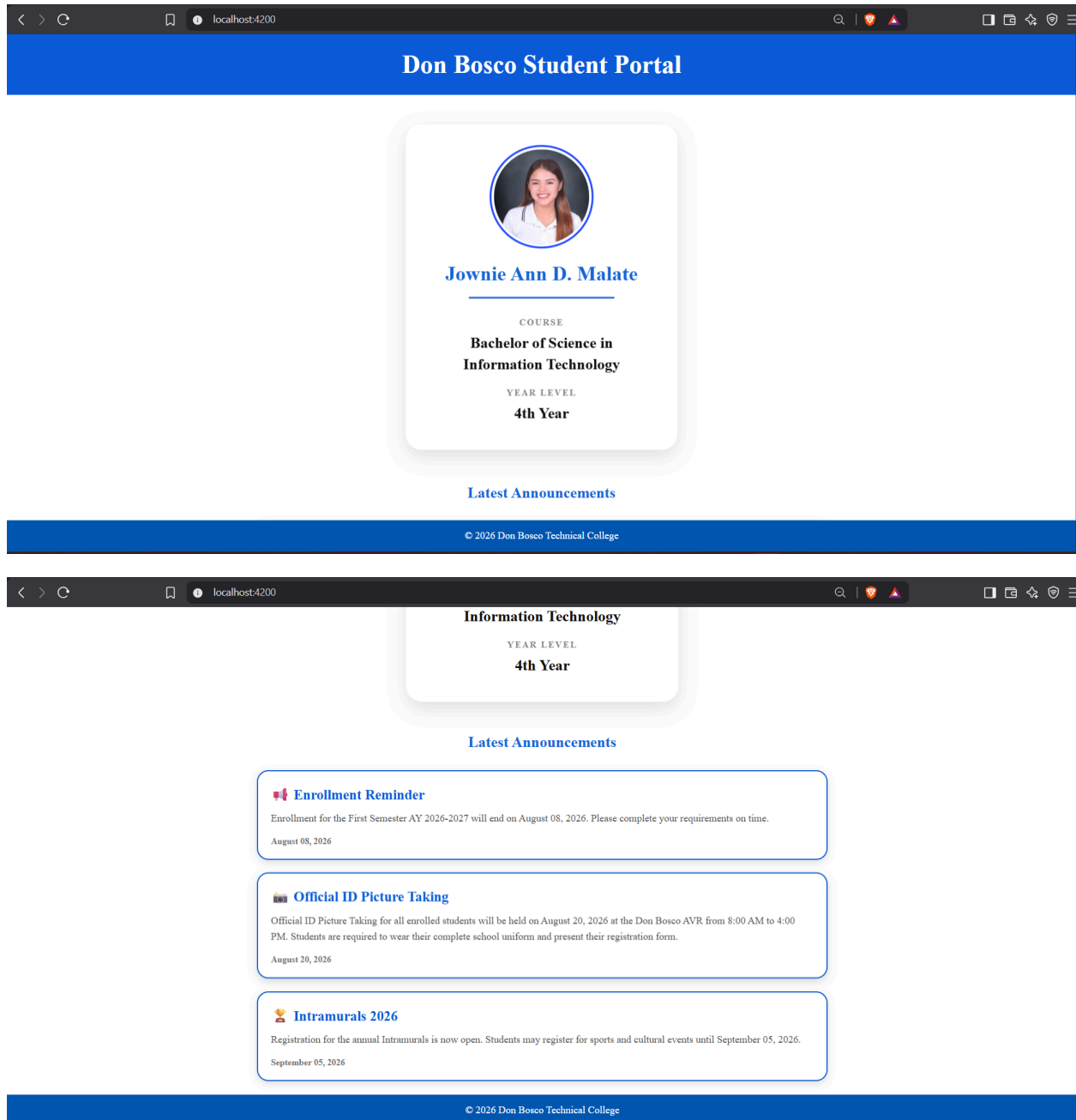
Evidence

- Browser before and after the CSS change
- `Announcement-card.css` code

BEFORE:



AFTER:



Flag 11 – Pushing Changes

Commit your completed work, then push to your repository.

 **Evidence**

- Terminal commands for committing + pushing
- Screenshot of the Github Repository reflecting the pushed changes

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Component update sent to client(s).
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git status
On branch feature/components
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   src/app/app.css
        modified:   src/app/app.html
        modified:   src/app/app.ts
        modified:   src/styles.css

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        public/profile.jpg
        src/app/components/

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git add
Nothing specified, nothing added.
hint: Maybe you wanted to say 'git add .'
hint: Disable this message with "git config set advice.addEmptyPathsSpec false"
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git add .

warning: in the working copy of 'src/app/components/announcement-card/announcement-card.html', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/app/components/announcement-card/announcement-card.spec.ts', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/app/components/announcement-card/announcement-card.ts', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/app/components/footer/footer.html', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/app/components/footer/footer.spec.ts', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/app/components/footer/footer.ts', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/app/components/header/header.html', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/app/components/header/header.spec.ts', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/app/components/header/header.ts', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/app/components/profile-card/profile-card.html', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/app/components/profile-card/profile-card.spec.ts', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/app/components/profile-card/profile-card.ts', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git commit -m "Complete Flags 4 to 10 - Angular reusable components"

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git commit -m "Complete Flags 4 to 10 - Angular reusable components"
>>
create mode 100644 src/app/components/announcement-card/announcement-card.html
create mode 100644 src/app/components/announcement-card/announcement-card.spec.ts
create mode 100644 src/app/components/announcement-card/announcement-card.ts
create mode 100644 src/app/components/footer/footer.css
create mode 100644 src/app/components/footer/footer.html
create mode 100644 src/app/components/footer/footer.spec.ts
create mode 100644 src/app/components/footer/footer.ts
create mode 100644 src/app/components/header/header.css
create mode 100644 src/app/components/header/header.html
create mode 100644 src/app/components/header/header.spec.ts
create mode 100644 src/app/components/header/header.ts
create mode 100644 src/app/components/profile-card/profile-card.css
create mode 100644 src/app/components/profile-card/profile-card.html
create mode 100644 src/app/components/profile-card/profile-card.spec.ts
create mode 100644 src/app/components/profile-card/profile-card.ts
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git push -u origin feature/components
Enumerating objects: 39, done.
Counting objects: 100% (39/39), done.
Delta compression using up to 8 threads
Compressing objects: 100% (31/31), done.
Writing objects: 100% (31/31), 156.23 KiB | 6.01 MiB/s, done.
Total 31 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 1 local object.
remote:
remote: Create a pull request for 'feature/components' on GitHub by visiting:
remote:   https://github.com/Jownieeey09/ITFE403A_Malate/pull/new/feature/components
remote:
To https://github.com/Jownieeey09/ITFE403A_Malate.git
 * [new branch]      feature/components -> feature/components
branch 'feature/components' set up to track 'origin/feature/components'.
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git branch
* feature/components
  main
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project>

```

Branches

New branch

Overview Yours Active Stale All

Default

Branch	Updated	Check status	Behind	Ahead	Pull request
main	3 days ago			Default	...

Your branches

Branch	Updated	Check status	Behind	Ahead	Pull request
feature/components	3 minutes ago		0	1	...

Active branches

Branch	Updated	Check status	Behind	Ahead	Pull request
feature/components	3 minutes ago		0	1	...